

Spoken Tutorial – Code and Audio Generation: The Specialists

Title: Code and Audio Generation: The Specialists

Module	Generative AI
Episode	Code and Audio Generation
Learning Objective	At the end of this tutorial learner will be able to 1. Understand the role of generative AI in code generation. 2. Identify specific tools and applications of AI in programming. 3. Explore how generative AI is used for audio creation. 4. Recognize the various applications of voice synthesis and music generation.
Approx. Duration	7 minutes
Outline	• Welcome to Code and Audio Generation • Learning Objectives • System Requirements • Prerequisites • Generative AI Beyond Text and Images • Code Generation: The Developer's Assistant • Introducing GitHub Copilot • How AI Assists in Coding: Code Suggestions • How AI Assists in Coding: Comments to Code • How AI Assists in Coding: Bug Finding • Benefits of AI in Software Development • Audio Generation: The Sound Wizards • Voice Synthesis and Text-to-Speech • Applications of Voice Synthesis • AI in Music Creation • Summary of Key Points • Assignment: Explore AI Tools • Thank You and Acknowledgements
Meta Tags	Generative AI, Code Generation, Audio Generation, GitHub Copilot, Voice Synthesis, Text-to-Speech, AI Music, Developer Tools, Sound Design
Pre-requisite Tutorial	Basic understanding of Generative AI concepts and their applications in text and image generation.

Script

Visual Cue	Narration
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Abstract digital background with glowing lines representing code and sound waves, with a friendly AI assistant character gesturing welcome.	Welcome to this Spoken Tutorial. It's on Code and Audio Generation. We will explore advanced AI applications. Specifically, code and audio generation. AI is making big strides here.
A list of bullet points with icons representing learning, code, audio, and applications, on a clean digital background.	In this tutorial, you will learn to: Understand AI in code generation. Identify AI coding tools. Explore AI in audio creation. Recognize voice synthesis uses. And AI music generation.
A laptop screen displaying a web browser, with Wi-Fi signal icon, and a mouse.	To follow this tutorial, you need a computer. It needs a web browser. An active internet connection is a must. No software installations are needed. This is a conceptual overview.
A stack of books or building blocks, with the top block labeled 'Code and Audio AI' and the bottom blocks labeled 'Text AI' and 'Image AI'.	You should know basic Generative AI. Familiarity with text and image AI helps. This tutorial builds on those topics.
A futuristic cityscape with glowing lines extending from text and image icons to code symbols and musical notes, showing expansion.	We've seen AI create text and images. But AI can do more. Its capabilities extend further. AI now masters complex areas. Like generating code. And realistic audio. Think about it.
A programmer sitting at a desk, with a holographic AI assistant floating beside them, showing lines of code on a screen.	Let's start with code generation. Imagine a skilled assistant. For developers. AI tools learn from public code. They enhance programming. Significantly.
The GitHub Copilot logo prominently displayed, with lines of code flowing around it in a digital environment.	Meet GitHub Copilot. It's a great example. It's an AI model. Trained on billions of code lines. It integrates into your coding environment. It's a smart programming partner.
A code editor interface showing a developer typing, with a greyed-out suggestion box appearing automatically with code snippets.	How does Copilot help? As you type, it suggests code. The next few lines. This saves time. Developers code faster. And more accurately.

A split screen showing a natural language comment on one side, and the corresponding generated code on the other side, with an arrow connecting them.	AI also translates comments. Into functional code. Write a comment. Describe your goal. Copilot generates the code. It links language and programming.
A magnifying glass hovering over lines of code, highlighting a red 'bug' icon, with a green 'fix' suggestion appearing.	AI helps find bugs too. It analyzes your code. For errors. It suggests fixes. And improvements. Debugging becomes efficient.
A developer smiling and working efficiently, surrounded by glowing graphs and symbols representing increased productivity and innovation.	AI tools don't replace humans. They empower programmers. Developers get faster. They become more efficient. They focus on big problems. And innovation.
An abstract representation of sound waves transforming into musical notes and speech bubbles, with a magical, wizard-like glow.	Now, to audio generation. These models are sound wizards. They learn from sounds. Voices and music. They create new audio content. Amazing, isn't it?
A microphone transforming text into realistic human-like sound waves, with diverse voice icons (male, female, child) around it.	Voice synthesis is key. Also called text-to-speech. AI creates realistic voices. Human-like voices. Customize tones. And emotions.
Icons representing an audiobook, a podcast microphone, and a virtual assistant (like a smart speaker), all connected by sound waves.	Voice synthesis has many uses. Audiobooks use it. Reading becomes accessible. AI narrates podcasts. Virtual assistants use it. Like Siri and Alexa.
A musical staff with notes being generated by an AI brain icon, surrounded by various musical instruments and a video camera icon.	AI also makes music. It's a revolution. Compose royalty-free music. For your videos. Generate new melodies. And harmonies. Even create songs like famous artists.

A concise infographic summarizing code generation and audio generation, with key benefits and examples for each, on a divided screen.	To summarize. AI goes beyond text and images. Code tools like Copilot help developers. They suggest code. Translate comments. And find bugs. Audio AI creates voices. And music. These tools boost creativity. And efficiency.
A person sitting at a computer, researching, with various AI tool logos floating around them, and a thought bubble showing exploration.	Now, for your assignment. Research an AI code tool. Or an AI audio platform. Explore its features. See how it works. Share your findings. In the comments.
The EduPyramids and SINE, IIT Bombay logos prominently displayed, with a 'Thank You' message and a subtle background of digital pyramids and circuits.	This Spoken Tutorial is from EduPyramids. And SINE, IIT Bombay. Thank you for joining. We hope you learned a lot. And felt inspired. Visit our website for more.